

# Farhan Fuad Abir

(+1) 689-270-1830  
farhan.fuad@ucf.edu  
Orlando, FL, US

[github.com/farhanfuadabir](https://github.com/farhanfuadabir)  
[linkedin.com/in/farhanfuadabir](https://www.linkedin.com/in/farhanfuadabir)

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| ACADEMIC              | <b>University of Central Florida</b> , Florida, US                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Aug 2024 - Present  |
| QUALIFICATIONS        | <i>Doctor of Philosophy</i> in Computer Engineering<br><i>Supervisor:</i> Dr. Laura Brattain<br><i>Area:</i> Working on ultrasound-based disease diagnosis with multimodal AI, exploring trustworthiness with explainable AI and exploring generative AI-based solutions to class imbalance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |
|                       | <b>University of Central Florida</b> , Florida, US                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Aug 2022 - Aug 2024 |
|                       | <i>Master of Science</i> in Computer Engineering<br><i>Supervisor:</i> Dr. Rickard Ewetz<br><i>Thesis:</i> <a href="#">Neuro-Symbolic Distillation of Reinforcement Learning Agents</a><br>- Developed an LLM-assisted neuro-symbolic feature extraction method for visual observation space<br>- Proposed a neural-guided symbolic model extraction method for continuous observation space                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CGPA: 3.89/4.00     |
|                       | <b>University of Dhaka</b> , Dhaka, Bangladesh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2016 - 2020         |
|                       | <i>Bachelor of Science</i> in Electrical and Electronic Engineering<br><i>Supervisor:</i> Dr. Mosabber Uddin Ahmed<br><i>Project:</i> <a href="#">A Framework for Sensor-Based Sign Language Detection using Machine Learning</a><br>- Developed a low-cost dataglove to detect 14 static and 3 dynamic hand gestures in real-time<br>- Made two datasets of static and dynamic gestures of 35 subjects using flex and IMU sensors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CGPA: 3.55/4.00     |
| SELECTED PUBLICATIONS | <b>Journal Articles</b> <ul style="list-style-type: none"><li>• <b>F. F. Abir</b>, M. E. H. Chowdhury, M. I. Tapotee, et al., "<a href="#">PCovNet+: A CNN-VAE anomaly detection framework with LSTM embeddings for smartwatch-based COVID-19 detection</a>", in <i>Engineering Applications of Artificial Intelligence</i> [IF - 7.802 (2023)], 2023, 106130</li><li>• S. Mahmud, N. Ibtehaz, A. J. Gonzales, T. Rahman, M. S. Hossain, M. S. A. Hossain M. A. A. Faisal, <b>F. F. Abir</b>, et al., "<a href="#">NABNet: A Nested Attention-guided BiConvLSTM network for a robust prediction of Blood Pressure components from reconstructed Arterial Blood Pressure waveforms using PPG and ECG signals</a>", in <i>Biomedical Signal Processing and Control</i> [IF - 5.076 (2023)], 2022, 104247</li><li>• M. N. I. Shuzan, M. E. H. Chowdhury, M. B. I. Reaz, A. Khandakar, <b>F. F. Abir</b>, et al., "<a href="#">Machine learning-based classification of healthy and impaired gaits using 3D-GRF signals</a>", in <i>Biomedical Signal Processing and Control</i> [IF - 5.076 (2023)], 2023, 104448</li><li>• <b>F. F. Abir</b>, K. Alyafei, M.E.H. Chowdhury, A. Khandakar, R. Ahmed, et al. "<a href="#">PCovNet: A presymptomatic COVID-19 detection framework using deep learning model using wearables data</a>", in <i>Computers in Biology and Medicine</i>, 2022, 105682 [IF - 6.69 (2022)]</li><li>• M. A. A. Faisal, <b>F. F. Abir</b>, M. U. Ahmed, M. E. H. Chowdhury, "<a href="#">Exploiting domain transformation and deep learning for hand gesture recognition using a low-cost dataglove</a>", in <i>Scientific Reports</i> [IF - 4.997 (2022)], 2022, 21446</li><li>• K. Alyafei, R. Ahmed, <b>F. F. Abir</b>, M. E. H. Chowdhury, et al., "<a href="#">A comprehensive review of COVID-19 detection techniques: From laboratory systems to wearable devices</a>", in <i>Computers in Biology and Medicine</i> [IF - 6.698 (2022)], 2022, 106070</li><li>• T. Rahman, A. Khandakar, <b>F. F. Abir</b>, M. A. A. Faisal, M. S. Hossain, et al., "<a href="#">QCovSML: A reliable COVID-19 detection system using CBC biomarkers by a stacking machine learning model</a>", in <i>Computers in Biology and Medicine</i>, 2022, 105284 [IF - 6.69 (2022)]</li><li>• M. M. Soliman, M. E. H. Chowdhury, M. T. Islam, F. Musharavati, M. Nabil, M. Hafizh, A. Khandakar, S. Mahmud, E. Z. Nezhad, M. N. I. Shuzan, <b>F. F. Abir</b>, "<a href="#">A Review of Biomaterials and Associated Performance Metrics Analysis in Pre-Clinical Finite Element Model and in Implementation Stages for Total Hip Implant System</a>", in <i>Polymers</i> [IF - 4.967 (2022)], 2022, 14, 4308</li></ul> |                     |

## Book Chapter

- **F. F. Abir**, M. A. A. Faisal, O. Shahid, M. U. Ahmed, "[Contactless Human Activity Analysis: an overview of different modalities](#)", in *Contactless Human Activity Analysis*, Springer Nature Switzerland AG, 2021, vol. 200, pp. 83-112.

## Conference Proceedings

- **F. F. Abir**, A. E. Daly, K. Anderman, T. Ozmen, & L. J. Brattain, "[Multimodal Deep Learning for Phyllodes Tumor Classification from Ultrasound and Clinical Data](#)", in *21st IEEE BSN'25*, 2025.
- M. A. A. Faisal, **F. F. Abir**, M. U. Ahmed, "[Sensor Dataglove for Real-time Static and Dynamic Hand Gesture Recognition](#)", in *10th ICIEV'21*, 2021, pp. 1-7. **[Best Paper]**
- M. S. Siraj, M. A. A. Faisal, O. Shahid, **F. F. Abir**, T. Hossain, S. Inoue, and M. A. R. Ahad, "[UPIC: User and position independent classical approach for locomotion and transportation modes recognition](#)", in *Adjunct Proceedings of the UbiComp-ISWC'20*, 2020, pp. 340–345.
- M. A. A. Faisal, M. S. Siraj, M. T. Abdullah, O. Shahid, **F. F. Abir**, and M. Ahad, "[A pragmatic signal processing approach for nurse care activity recognition using classical machine learning](#)", in *Adjunct Proceedings of the UbiComp-ISWC'20*, 2020, pp. 396–401.

|                             |                                                                                                                                |                     |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------|
| ATTENDED<br>CONFERENCES     | IEEE-EMBS International Conference on Body Sensor Networks 2025                                                                | Nov 2025            |
|                             | IEEE CS BDS Winter Symposium'20 [Online]                                                                                       | Nov 2020            |
|                             | <i>Presentation: Locomotion and Transportation Modes Recognition Using Smart Feature based Machine Learning</i>                |                     |
|                             | UbiComp-ISWC'20 [Online]                                                                                                       | Sep 2020            |
|                             | ICIEV-IVPR-ABC'20 [Online]                                                                                                     | Aug 2020            |
| INVITED<br>TALKS            | <a href="#">PCB Design 101</a>                                                                                                 | Aug 2020            |
|                             | - Designed the 10-days online course curriculum delivered via Microsoft Teams and managed with Google Classroom                |                     |
|                             | - Introduced the basic steps of Circuit Schematic and PCB designing with Proteus Design Suite                                  |                     |
|                             | - Taught design techniques to 30 participants hands-on in 3 live sessions                                                      |                     |
|                             | <a href="#">Discussion on Preliminary Ideas of Machine Learning</a> , IEEE Student Branch DU                                   | Sep 2019            |
|                             | - Presented the scopes of using ML for Hand Gesture Recognition to 45 sophomore students                                       |                     |
| PROFESSIONAL<br>EXPERIENCES | <a href="#">Introduction to Automation</a> , IEEE RAS DU                                                                       | Jul 2018            |
|                             | - Introduced the locally available embedded systems modules for automation to 23 freshman students                             |                     |
|                             | <b>Graduate Research Assistant</b> , Dr. Laura Brattain                                                                        | Aug 2024 - Present  |
|                             | <i>Project: Multimodal deep learning model for ultrasound-based disease diagnosis leveraging clinical data</i>                 |                     |
|                             | - Used the method for breast cancer and pleural effusion diagnosis                                                             |                     |
|                             | - Implemented feature-level fusion for ultrasound image and clinical data                                                      |                     |
|                             | - Explored CAM-based explainability approaches for diagnosis validation                                                        |                     |
|                             | <i>Project: Hybrid diffusion model for high-fidelity breast ultrasound image generation</i>                                    |                     |
|                             | - Used both text-to-image and image-to-image stable-diffusion models to generate breast ultrasound images of different classes |                     |
|                             | <i>Project: Ultrasound-based Neck-landmark Detection</i>                                                                       |                     |
|                             | - Implemented YOLOv11 for object detection with class-aware sampling to mitigate class imbalance                               |                     |
|                             | <b>Graduate Research Assistant</b> , Dr. Rickard Ewetz                                                                         | Oct 2022 - Aug 2024 |
|                             | <i>Project: Neurosymbolic Reinforcement Learning</i>                                                                           |                     |
|                             | - Explored the potential of using neurosymbolic approach for interpretable reinforcement learning agents                       |                     |
|                             | - Mentored 2 undergraduate students on interpretable reinforcement learning                                                    |                     |
|                             | <b>Research Assistant</b> , Qatar University Machine Learning Group                                                            | Aug 2021 - Aug 2022 |
|                             | <i>Project: Presymptomatic COVID-19 Detection Using Smartwatch Data</i>                                                        |                     |
|                             | - Worked with 3 biomarkers derived from smartwatch data to extract underlying diseases                                         |                     |

- Experimented on deep learning-based anomaly detection models
- Project: IoT-Enabled Automatic IV Dispenser System for Burn Patients*
- Designed on 3 subsections - saline drip detector, automated saline flow controller, and urine output monitor systems
- Integrated the systems in the IoT backend for real-time monitoring and control

**Biomedical Engineer**, Datasoft Manufacturing & Assembly Inc. Jul 2021 - Sep 2021

*Project: Virtual Doctor - A IoT-enabled Telemedicine Platform*

- Developed three devices to monitor vitals – Heart Rate, Blood Pressure, Oxygen Saturation, and Temperature
- Integrated the devices in the telemedicine platform for real-time monitoring
- Coordinated a team of six interns

**Embedded System Engineer**, Quanfield Technologies Jun 2016 - Jan 2018

*Project: ArcherClock* - A timer and scoring system for archery competition

- Developed the system protocol, designed the PCB of the whole system, and used Raspberry Pi to control and handshake with the mobile app

*Project: QuizzoBuzz* - A buzzer system for quiz competition

- Designed the circuits and PCB of six buzzer terminals and programmed the AVR microcontroller

STANDARDIZED **GRE** - 320 (Quant: 164, Verbal: 156, AWA: 4.0) Oct 2020

TEST SCORES **IELTS** - 7.5 (Listening: 8.5, Reading: 7.5, Writing: 6.5, Speaking: 7) Oct 2021

RELEVANT *Languages: C/C++, Python (PyTorch, TensorFlow, Keras, SciKit Learn, Django)*

SKILLS *Web: HTML, CSS, Bootstrap*

*Others: Matlab, Circuit Simulation and Prototyping (LTSpice), PCB Design (Proteus, Eagle), PCB Fabrication*

ADDITIONAL **ASL Dataglove** - A sensor-based dataglove to interpret 40 ASL gestures 2021

PROJECTS **Tothastu** - A blog created with Django 2021

**BioAI** - AI-based medical assistant application based on Google BERT 2020

**Pet Robot** - Companion robot capable of following a certain color tag 2018

**Line Follower Robot** - A robot with path following ability under various challenges 2016

HONORS & **ORCGS Doctoral Fellowship**, University of Central Florida, United States Aug 2022

AWARDS **Best Paper**, Int'l. Conference on Informatics, Electronics & Vision 2021, Japan Aug 2021

**Champion**, Satellite Expedition Idea Contest, Brac University, Bangladesh Jan 2021

**Runner-up**, Idea Contest, 1st National Workshop on BDML, Bangladesh Dec 2020

**6<sup>th</sup> Position**, Sussex-Huawei Locomotion Challenge 2020, Cancún, Mexico Sep 2020

**Finalist**, APSCO Micro-satellite Idea Contest, Xi'an, China Oct 2019

**Runner-up**, Matlab Contest, IUT, Dhaka, Bangladesh Sep 2019

**Runner-up**, Project Showcasing, FabLab DU, Dhaka, Bangladesh Nov 2018

**Champion**, Line Follower Robot Competition, DUET, Dhaka, Bangladesh Jul 2017

VOLUNTEER **President**, Bangladeshi Student Association at UCF 2025 - Present

EXPERIENCES **Secretary and Treasurer**, IEEE Student Branch DU 2019 - 2021

**Volunteer**, FabLab University of Dhaka 2018

**Volunteer**, CommunityAction (UN-recognized Social Welfare Organization) 2016 - 2017